Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

Methodologies that were used:

- Data Collection with web scraping and Space X API.
- Exploratory Data Analysis (EDA), data wrangling.
- Data visualization and interactive.
- Visual analytics.
- Machine Learning Prediction.

Summary of all results

- The data were obtained from public sources.
- To predict the success of launching it was necessary to identify the best features, and for that, EDA was necessary.

Introduction

Space X Falcon 9 First Stage Landing Project

Capstone, predict if the Falcon 9 first stage will land successfully. SpaceX advertises Falcon 9 rocket launches on its website.

- Best way to estimate the total cost for launches, by predicting successful landings of the first stage of rockets;
- Find the best place to make successful launches.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Data extracted directly from SpaceX's database using their public API.
  - HTTP Requests: Python request library to send GET requests to SpaceX API endpoints as launches and past endpoints.
  - JSON: Convert the raw API into flat pandas format
  - Data retrieval: Make additional API calls to fetch the specific details for aeach ID, as the booster version, payload mass, orbit type etc...

Methodology

- Perform data wrangling

  Raw collected data needs to be clean it up.

  - Missing Values: identified and dealt with the missing data, calculating the mean to fill the blank space.
  - Filtering: To only include "Falcon 9" launches. Focus the machine learning training on the consistent rocket plaform.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Data Collection

- Data sets were collected from Space X API and from Wikipedia
  - Links: https://api.spacexdata.com/v4/rockets/
  - https://en.wikipedia.org/wiki/List_of_Falcon/_9/_and_Falcon_Heavy_launches

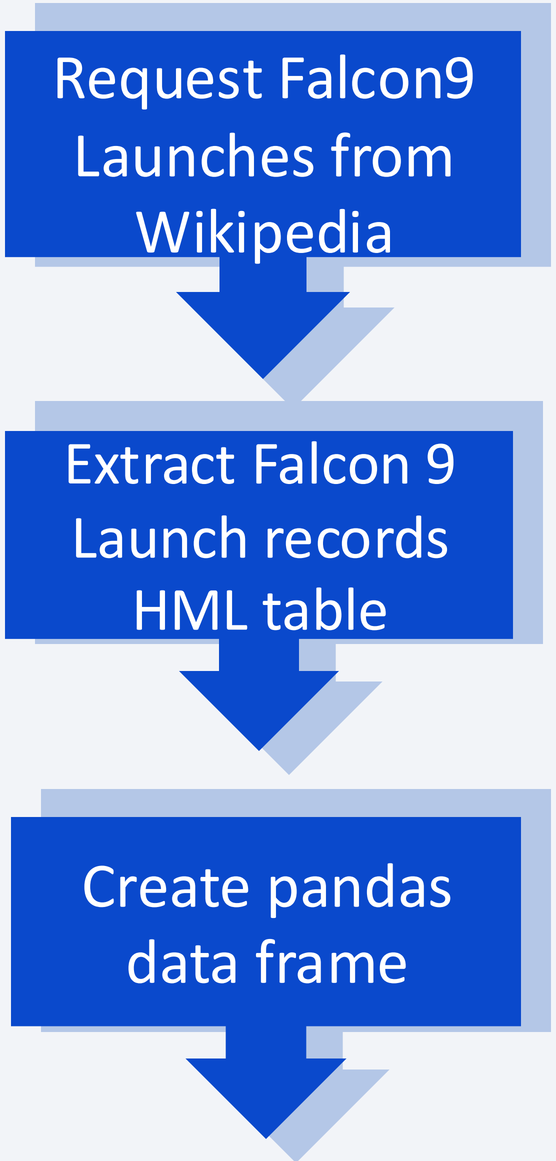
- Web scraping technics were used.

Data Collection – SpaceX API

- The data is extracted directly from SpaceX database using their public API.
- The API was used as the flowchart shows.
- Persisted data.
- GitHub Link

Data Collection - Scraping

- SpaceX Falcon 9 launches records are from Wikipedia page title: "List of Falcon 9 and Falcon Heavy launches".
- Scrap Falcon 9 launch records with "BeautifulSoup".
- The downloaded data was used as the flowchart shows.
- [GitHub Link]



Data Wrangling

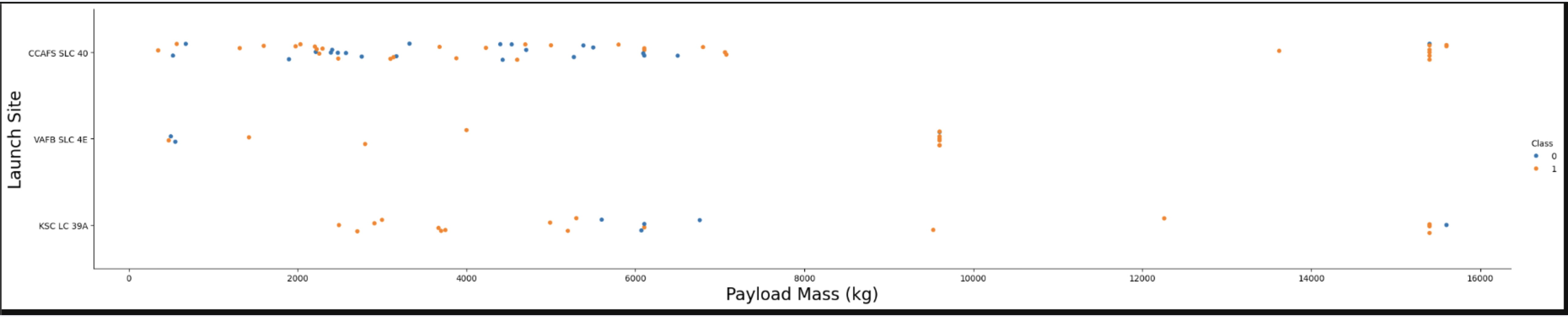
- EDA (Exploratory Data Analysis) was performed on the dataset. To find patterns on the data and determine the labels for training the model.
- Calculate the number of launches, launches per site, the orbit aimed for the launch and occurrences of mission outcome per orbit type.
- At last, from the Outcome column we create the label "landing outcome", the classification variable represent that outcome. If the value is zero means unsuccessfull and one success.
- GitHub Link

EDA → Summary → Creation of the column Landing Outcome

EDA with Data Visualization

- To visualize the relationships between the features we used scatterplotts and barplots were used:
  - Payload Mass X Flight Number, Launch Site X Flight Number, Launch Site X Payload Mass,Orbit and Flight Number, Payload and Orbit

- [GitHub Link]



EDA with SQL

SQL queries performed

- Names of the unique launch sites.
- Top 5 Launches sites, beginning with 'CCA' string.
- Total Playload Mass carryed by booster launched by NASA (CRS).
- Average Playload Mass carried by booster F9 v1.1
- Date of the first successful landing outcome in ground pad was achived.
- List the names of the boosters wich was success in drone ship and have payload mass greater than 4000 but less than 6000.
- List of the total number of successful and failure mission outcomes.
- List all the booster versions that have carried the maximum payload mass
- List of the failed landing outcomes in drone ship, booster version and launch site names for year 2015.
- Rank of the count of landing outcomes, such as droneship or the success ground pad
- GitHub Link

Build an Interactive Map with Folium

- Circle, lines and markerclusters were used with Folium Maps.
- It was expected to use the markers to indicate the points of launch site.
- The circles indicate the highlighted areas around specific coordinates, such as the NASA Johnson Space center.
- The marker clusters indicates groups of events in each coordinate, like launches in a launch site.
- The lines are used to indicate the distance between two coordinates.
- GitHub Link

Build a Dashboard with Plotly Dash

Plots and graphs

To visualize the next data:

- Percentage of launches by site
- Playload ranges

This allowed to analyze the relation between payload and launch sites and helping to identify the best place to launch

- GitHub Link

Predictive Analysis (Classification)

Classification models

- Four models were compared
  - Logistic regression
  - Support vector machine
  - Decision tree
  - K nearest neighbors

Data Preparation And standarization → Testing the models with each combinations of parameters → Results Comparison

[GitHub Link]

Results

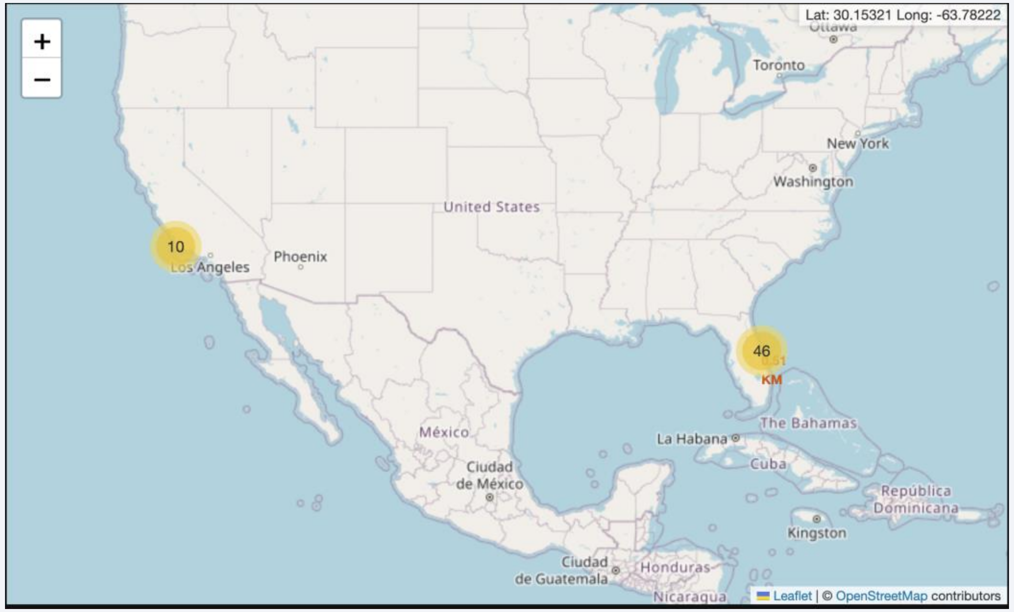
Exploratory data analysis results

- Space X uses 4 launch sites.
- The first launches were done by the NASA and Space X.
- The first success landing outcome happens in 2015, five years after the first launch.
- The average payload mass of F9 v1.1 booster is 2,928 kg.
- Falcon 9 booster versions were successful at landing in drone ships and having the payload above the average level.
- The number of landing outcomes increased over the years passed.
- Almost 100% of missions outcomes were sucessful.

Results

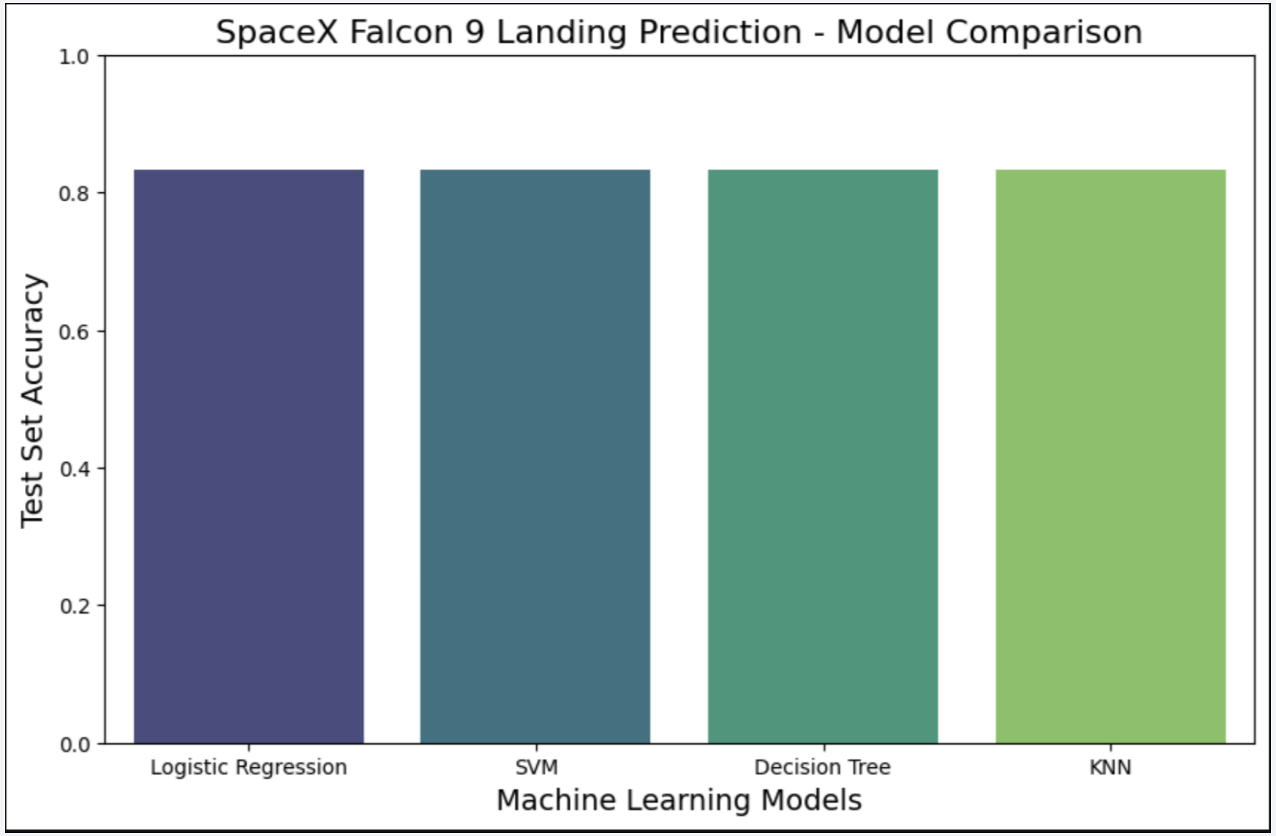
Interactive analytics demo in screenshots

- Using interactive analytics was possible to locate the launch sites.
- We can see that this places are positioned in safe places, such as being near the sea.
- Much of the launches were at the east cost site.



Results

Predictive analysis results

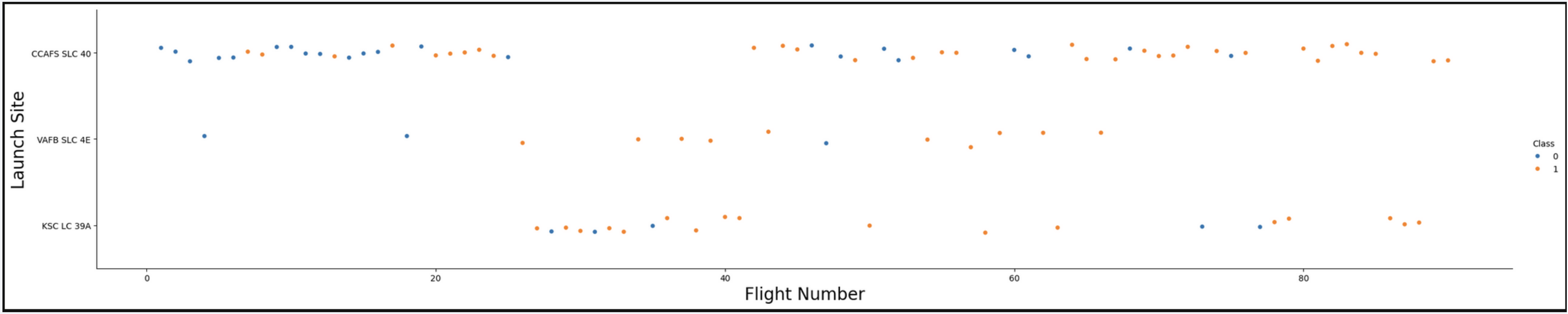


Section 2

Insights drawn from EDA

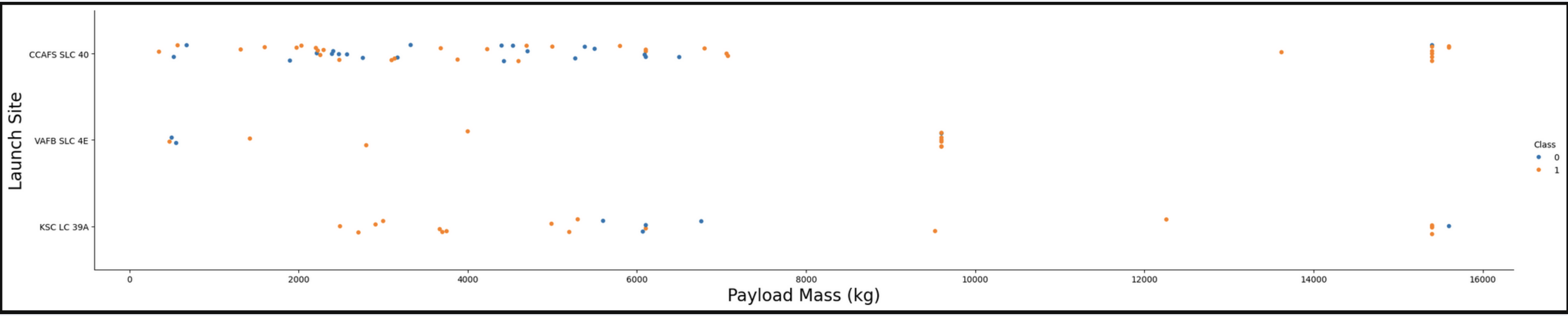
Flight Number vs. Launch Site

- The best launch site is the CCAF5 SLC 40.
- The second place VAFB SLC 4E and the third place is the KSC LC 39A.
- We can observe that the success rate increse over the time pass.



Payload vs. Launch Site

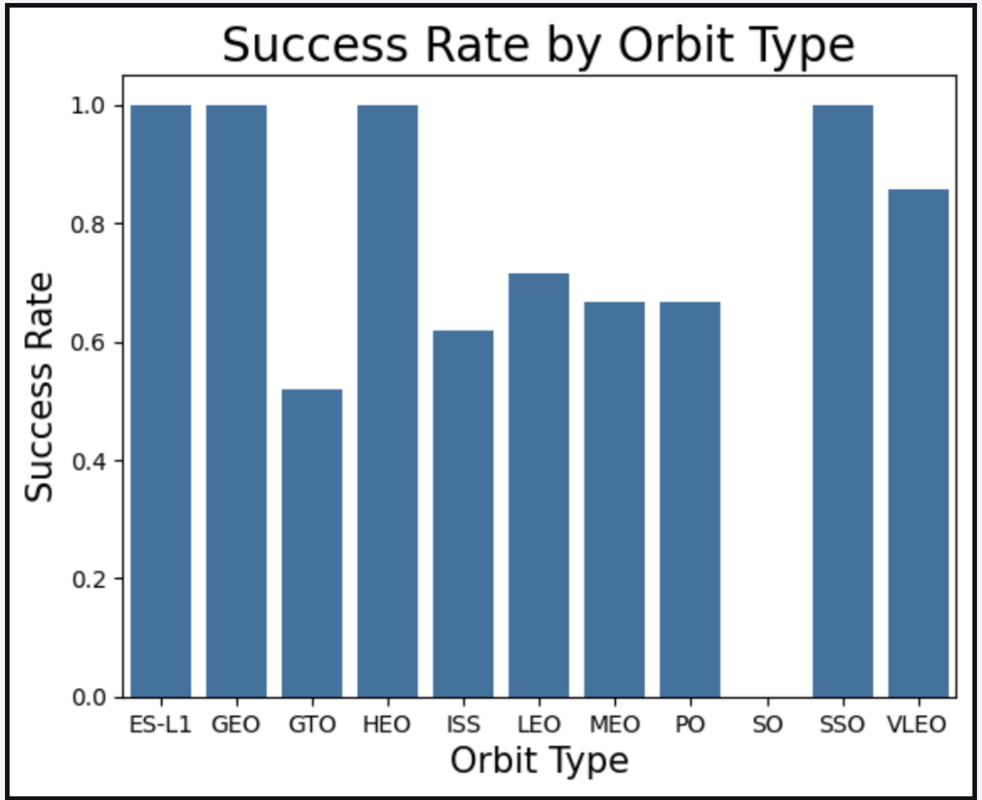
- We observe that the VAFB-SLC launchsite there are no rockets launched for heavy payload mass over 9000kg.
- Payloads over 10,000 is to be possible only on CCAFS SLC 40.



Success Rate vs. Orbit Type

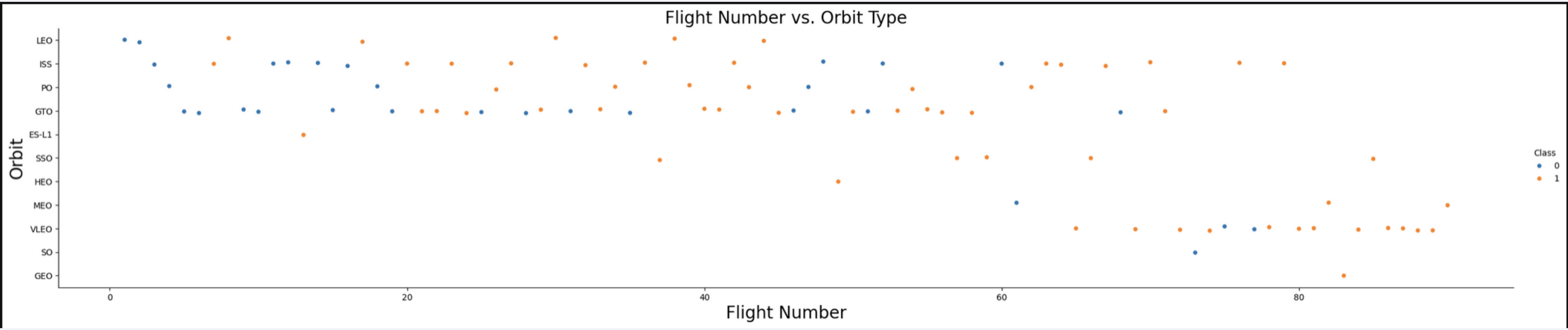
- The orbits has the biggest success rate:
  - ES-L1
  - GEO
  - HEO
  - SSO

- The orbits has the biggest success rate:
- Above 80% VLEO



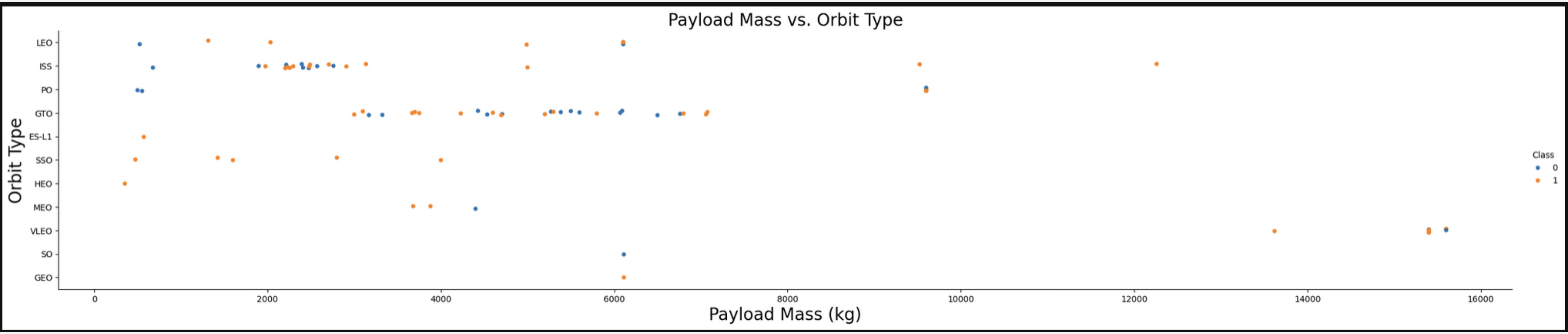
Flight Number vs. Orbit Type

- The success rate improvee over time in all orbits.
- VLEO orbit is the one that over time improve notoriously



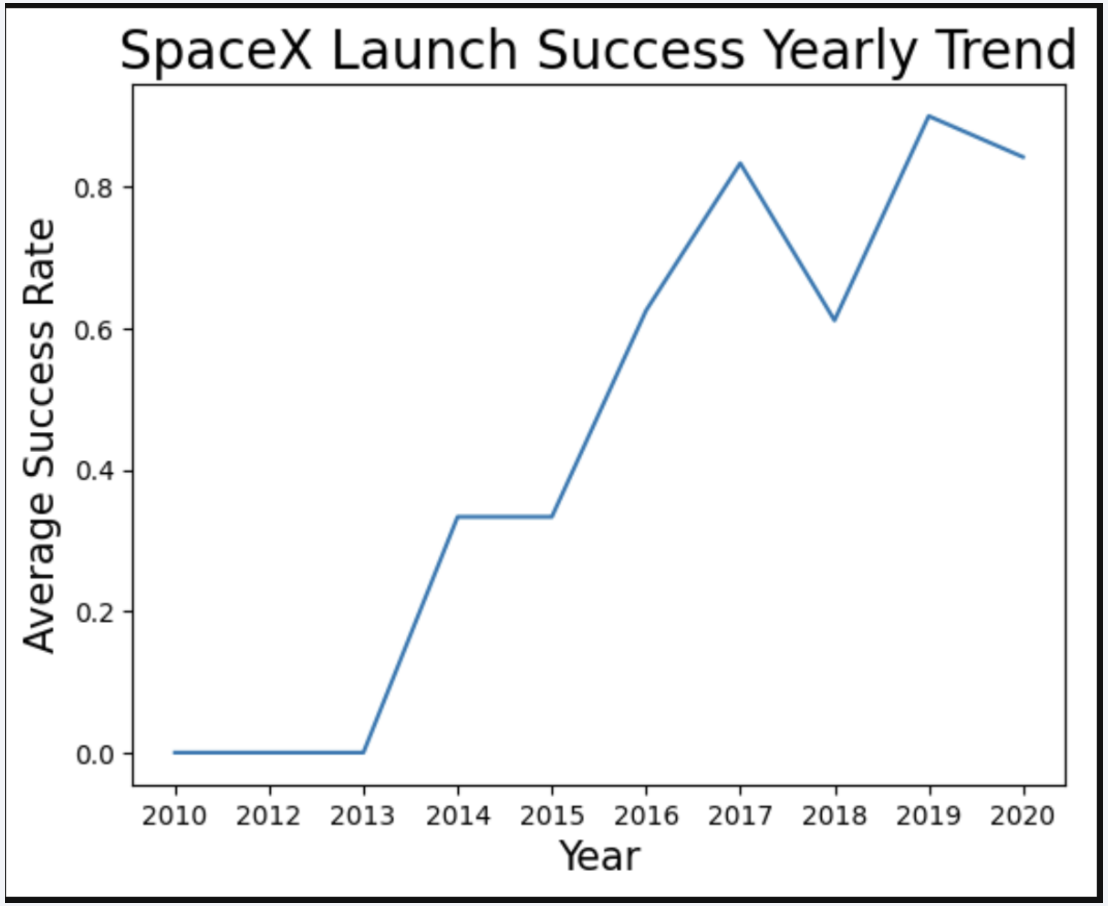
Payload vs. Orbit Type

- Successful landing rate on heavy payloads are more for PO, LEO and ISS.
- ISS orbit has the widest range



Launch Success Yearly Trend

- We can observe that the period of adjust were the first three years.
- The success rate started in 2013 and keep it forward until 2020.



All Launch Site Names

- The data shows that there are four launch sites.
- The query: %sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

	Launch Site
0	CCAFS LC-40
1	CCAFS SLC-40
2	KSC LC-39A
3	VAFB SLC-4E

Launch Site Names Begin with 'CCA'

- 5 records where launch sites begin with CCA
- The query: %sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Total payload carried by boosters from NASA
- The query: %sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
- The Total payload is calculated above by summing all payloads codes.

SUM(PAYLOAD_MASS__KG_)
45596

Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 v1.1
- The query: %sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
- Filter the data by the booster version.

AVG(PAYLOAD_MASS__KG_)
2928.4

First Successful Ground Landing Date

- Dates of the first successful landing outcome on ground pad:
- The query: %sql SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
- Selecting the booster versions according to the filters.

MIN(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- The query: %sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000 AND Landing_Outcome = 'Success (drone ship)';
- Selecting distinct booster versions according to the filters.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- The query: %sql SELECT Mission_Outcome, COUNT(*) as Total_Count FROM SPACEXTABLE GROUP BY Mission_Outcome;
- Mission outcomes and counting records for each group.

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- The query: %sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
- The boosters wich have the carried the maximum payload mass that are registered in the dataset

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4

Booster_Version
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- The query: %sql SELECT substr(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date, 1, 4) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
- Only two occurrences happened

Booster_Version	Launch_Site
F9 v1.1 B1012	CCAFS LC-40
F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

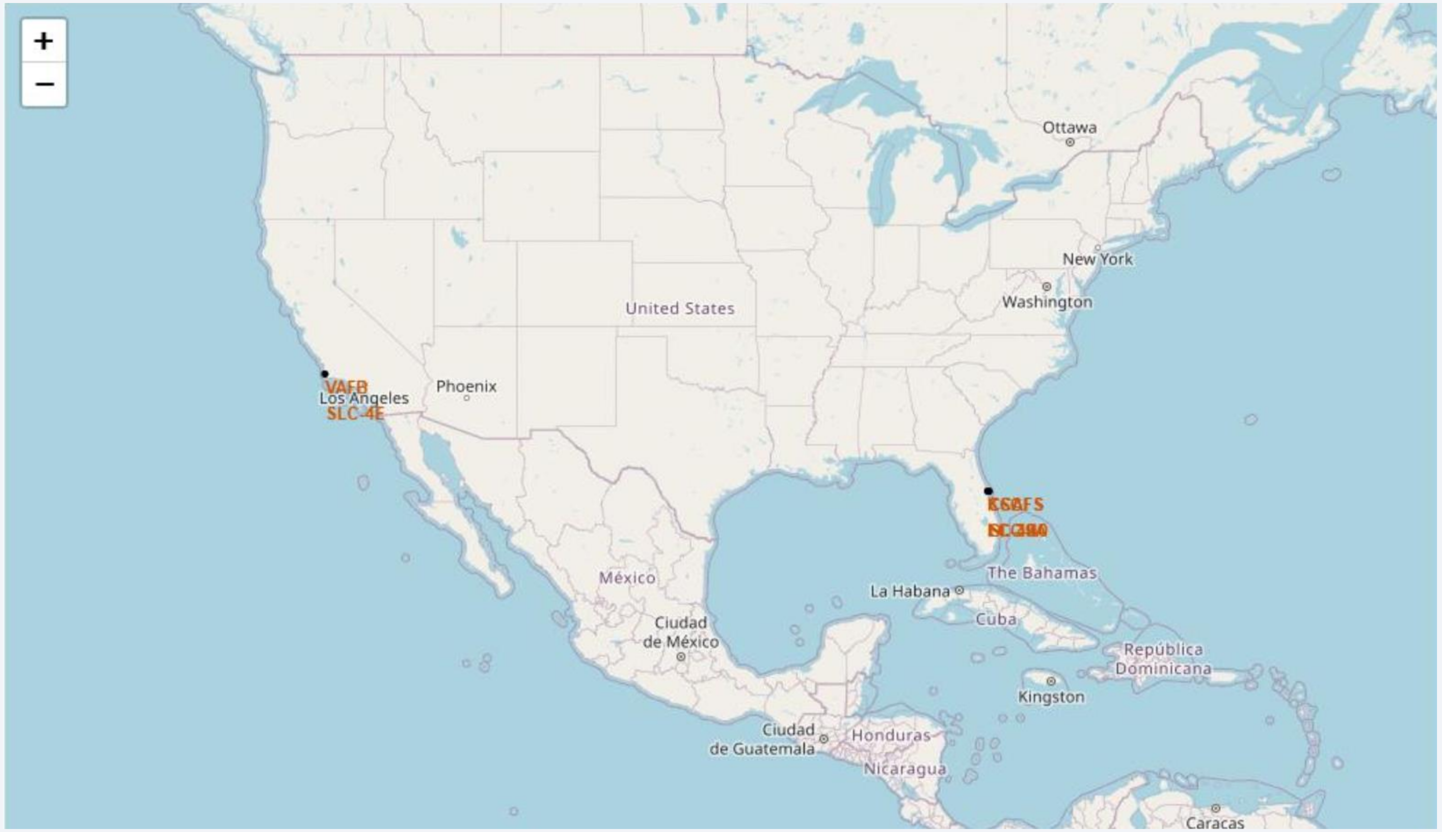
- Ranking of all landing outcomes between(the date 2010-06-04 and 2017-03-20.
- The query: %sql SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

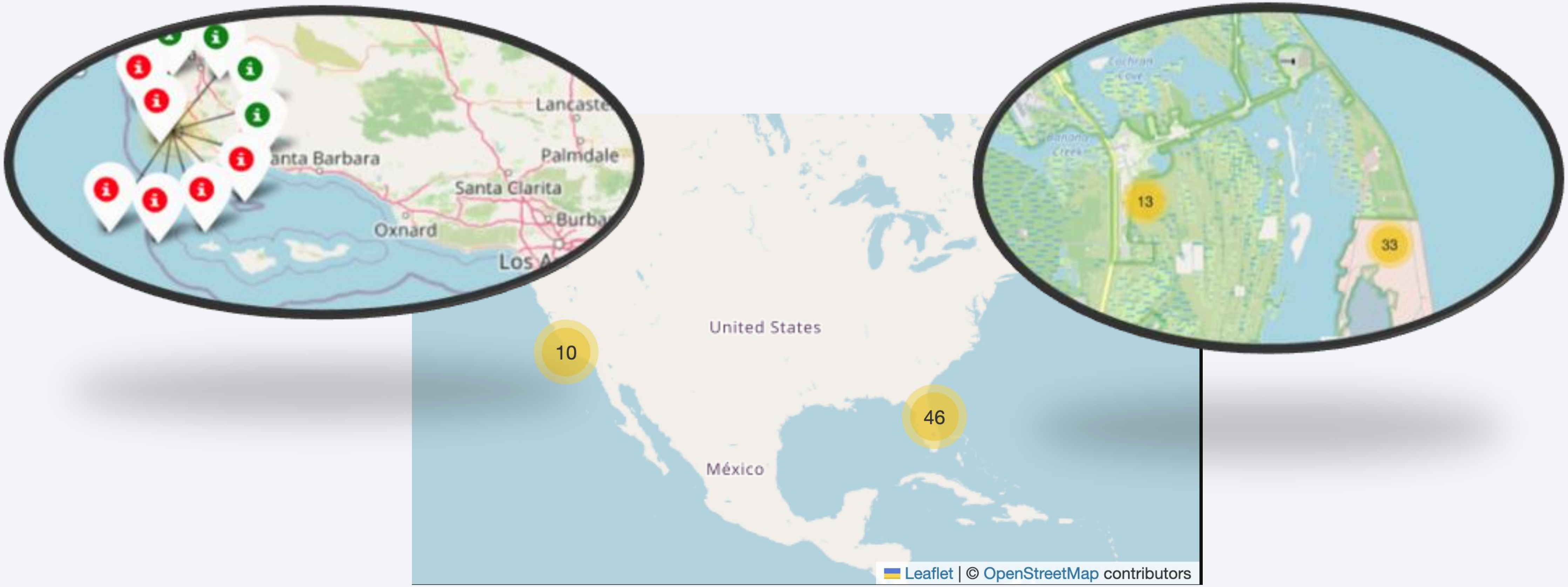
Launch Sites Proximities Analysis

All Launch Sites



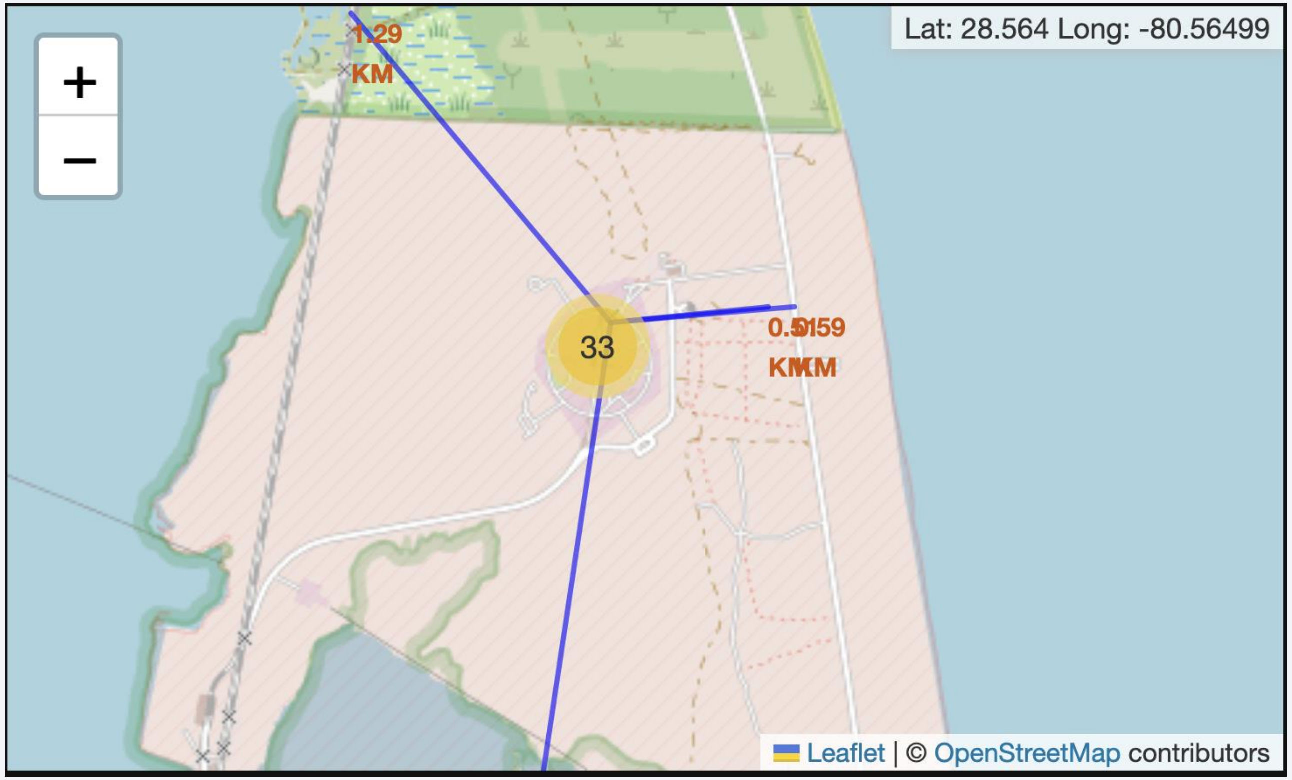
- The Launch sites are near the sea (probably for safe) and rail roads.

Launch Outcomes by Sites



The green markers represents the success and the red ones failures.

Safety logistic

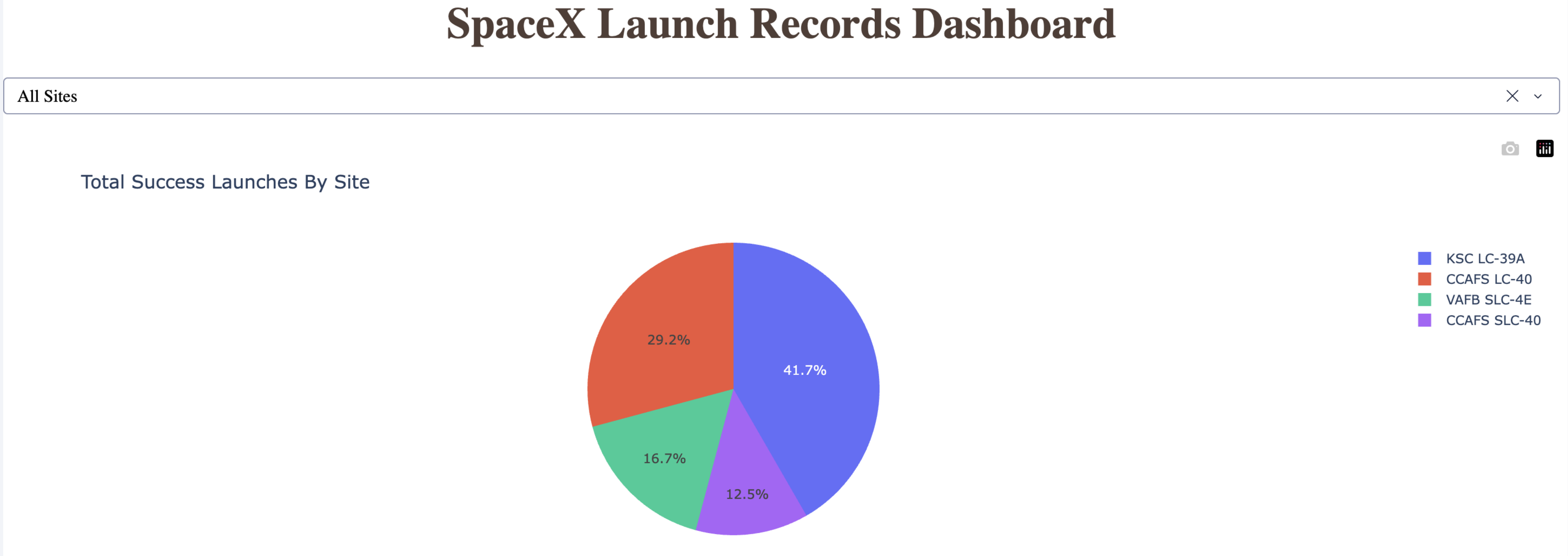


For a good launch site a good logistic is necessary, being near the sea, roads and relatively far from habitat areas.

Section 4

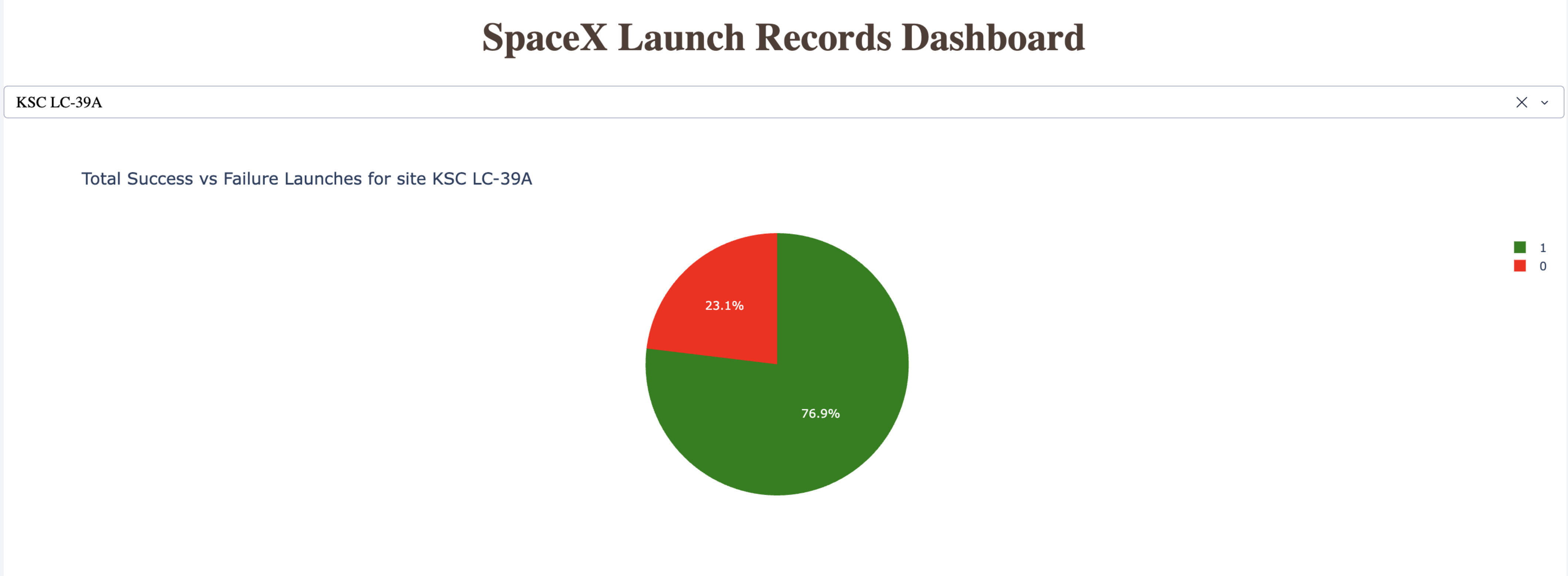
Build a Dashboard with Plotly Dash

Successful Launches



We can observe that the most important factor for the success of the missions is the place from which the launch is made.

Launch Success Ratio



- KSC LC 39A site, mission success 76.9% of the launches.

Payload vs Launch Outcome

The most successful combinations are the payloads under 6,000kg and FT booster.

Payload vs Launch Outcome

There is not enough data to determine the risk of launches over 7,000 kg

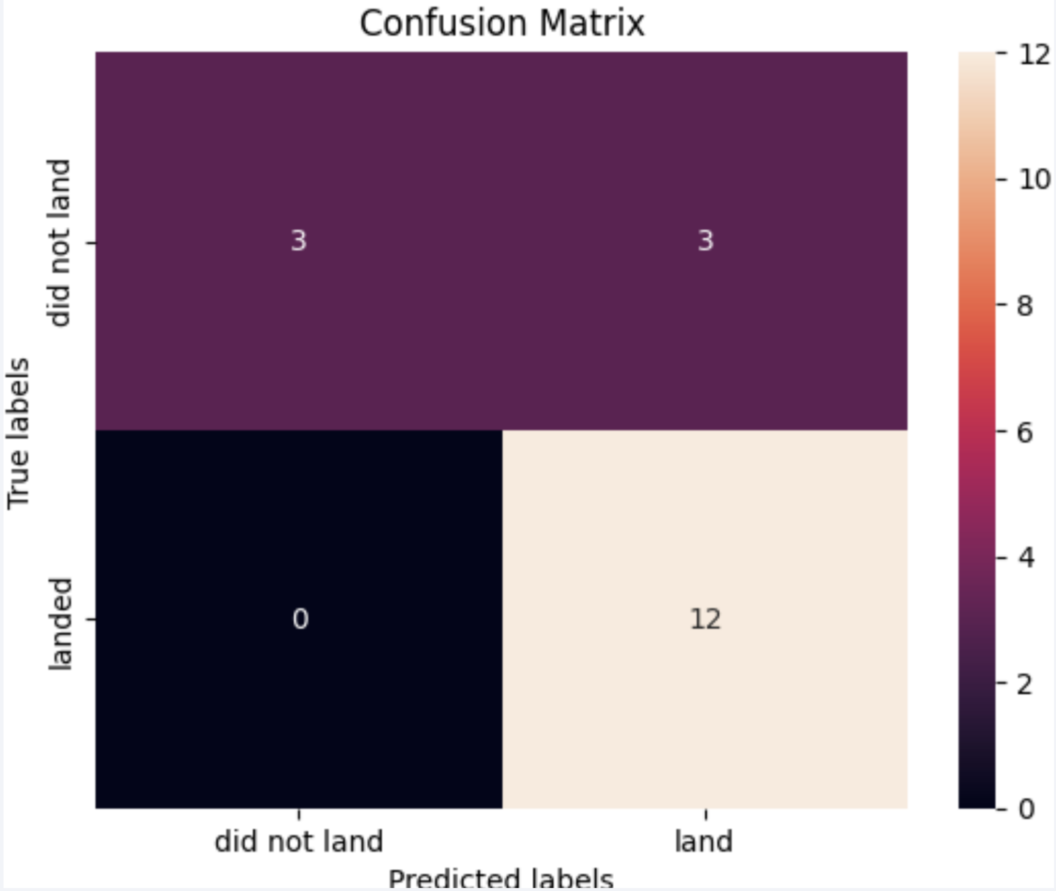
Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Four different classification models were tested.
- The model with the better accuracy classification is the Decision Tree Classifier.

Confusion Matrix



- Confusion matrixof Decision Tree Classifier proves its accuracy.

Conclusions

- The best Launch site is KSC LC-39A
- Launches under 6000 kg are less risk
- Decision Tree Classifier predicted more accurate the succesful landings.
- Most of the mission outcomes were successful over the time pass, that according of the evolution of the processes and the rockets.

Appendix

- I upload the cvs files that were generated and the screenshoots too.

Thank you!